INDIAN INSTITUTE OF TECHNOLOGY DELHI

New Delhi – 110016

INTERNSHIP REPORT

Arduino Based Embedded Systems

Submitted by

Ganesh Mahata

Entry No.:

Department of Biotechnology

Submitted To

Guide Name

Designation

Organization

Summer Internship 2026

Contents

List of Figures

List of Tables

Chapter 1

Introduction

This internship focused on the design and implementation of embedded systems using the Arduino UNO platform. The experiments cover basic digital interfacing, sensor integration, display modules, automation, and embedded applications.

The objective of the internship was to strengthen practical knowledge of microcontrollers, embedded programming, hardware interfacing, and real-time control systems through hands-on projects.

The report documents each experiment with its objective, circuit design, implementation methodology, working principle, results, and conclusion.

Chapter 2

Fundamentals of Arduino Interfacing

2.1 Experiment 0: LED Flash System

2.1.1 Project Name

LED Flash System

2.1.2 Problem Statement

The objective of this experiment is to interface an LED with Arduino UNO and make the LED blink continuously at a fixed interval.

This experiment helps in understanding:

- Digital output pins of Arduino
- Basic Arduino programming structure
- Use of functions in Arduino
- Timing control using `delay()`

2.1.3 Components Used

Table 2.1: Components Used

Component	**Quantity**
Arduino UNO	1
LED	1
100Ω Resistor	1
Breadboard	1
Jumper Wires	Required

2.1.4 Pin Connections

Table 2.2: Pin Connections

Device	**Arduino Pin**
LED Positive Terminal	D12
LED Negative Terminal	GND

2.1.5 Software Used

- Arduino IDE
- SimulIDE

2.1.6 Circuit Diagram

`Exp-0/led_flash/led_flash.png`

Figure 2.1: Circuit Diagram of LED Flash System

2.1.7 Source Code

The complete Arduino source code for this experiment is available in the project repository.

Click here to view the Arduino Source Code

2.1.8 Working Principle

1. The LED is connected to digital pin D12 of Arduino UNO.
2. In the `setup()` function, pin D12 is configured as an output pin using `pinMode()`.
3. The `loop()` function continuously calls the `toggle_led()` function.
4. Inside the `toggle_led()` function:
   - The LED turns ON using `digitalWrite(HIGH)`
   - Arduino waits for 1 second using `delay(1000)`
   - The LED turns OFF using `digitalWrite(LOW)`
   - Arduino again waits for 1 second
5. This process repeats continuously, causing the LED to blink repeatedly.

2.1.9 What I Learned

- Arduino UNO digital pins give approximately 5V output.
- Recommended current per digital pin is 20mA.
- Maximum current per digital pin is 40mA.
- Typical LED operating current is around 10mA to 20mA.
- LEDs should be connected using a current limiting resistor.
- Basic LED interfacing with Arduino UNO.

2.1.10 Conclusion

The LED Blinking System using Arduino UNO was successfully implemented and tested.

The LED blinked continuously with a delay of 1 second ON and 1 second OFF. This experiment helped in understanding basic Arduino programming, digital output control, and hardware interfacing.

2.1.11 References

1. Arduino Official Website
   https://www.arduino.cc/
2. Arduino UNO Documentation
   https://docs.arduino.cc/hardware/uno-rev3/

3. Arduino Tutorials
   https://www.arduino.cc/en/Tutorial/HomePage
4. Arduino IDE Documentation
   https://docs.arduino.cc/software/ide/
5. SimulIDE
   https://simulide.com/